Controlled numerical four-level counterexample to triplet screening of the Markovian Mpemba effect

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Local rate tests derived for three-level subsystems have been proposed as a scalable screen for the Markovian Mpemba effect in larger detailed-balance systems. We test the finite-temperature proposal in the minimal four-level setting and give a simple controlled-numerical witness. In units where the bath inverse temperature and a common rate scale are one, the energies are $(0, 1/200, 7/20, 14)$ and the six downward rates are $(1/10, 1, 13, 99/100, 1, 3/4)$. Every one of the four triplets strictly fails its applicable published necessary-condition predicate, while the slow nonstationary mode is simple. Nevertheless, Gibbs preparations at inverse temperatures $1/50$ and $1/20$ have normalized slow-mode amplitudes 0.0628307488472 and 0.0630538586374, respectively. Thus the initially hotter state has the smaller late-time amplitude, and direct relative-entropy evolution reverses the initial distance ordering. Separate generator constructions, 100- and 180-digit calculations, residual tests, and direct dynamics reproduce the effect. The result is a reproducible computational counterexample at controlled numerical scope, not a directed-rounding proof or a classification of the heating branch or of general N.

I. INTRODUCTION

The Mpemba effect denotes anomalous relaxation in which a preparation initially farther from a common equilibrium state becomes closer at a later time. Modern formulations separate this phenomenon from the material-specific question of freezing water and place it in the spectral theory of nonequilibrium relaxation [1]. For finite Markov systems obeying detailed balance, the overlap of a thermal preparation with the slowest nonstationary mode gives a particularly clean late-time criterion [2, 3].

This criterion is simple once the full generator has been diagonalized, but it does not by itself provide an inexpensive way to screen a large parameter space. Avitan, Factor, and Gelbwaser-Klimovsky recently derived elementary necessary conditions for a three-level system and proposed applying them to every triplet of an N-level generator [4]. They proved an exclusion statement at zero bath temperature under stronger inequalities and reported finite-temperature random tests, while the implication from triplet exclusion to global exclusion at finite temperature remained conjectural. A subsequent study found four-level Mpemba behavior even when no isolated triplet itself exhibits the effect [5]; that premise is weaker than failure of every published necessary-condition predicate. Computational screening has also been explored in other Markovian settings [6].

Here we address the stronger, unresolved premise directly. We exhibit one four-level, finite-positive-temperature, detailed-balance generator for which all four applicable triplet predicates fail strictly, yet two cooling preparations have the reversed slow-mode ordering. The exact input parameters are rational, but the Boltzmann factors and eigensystem are evaluated numerically. We therefore state the result at controlled-numerical scope and retain explicit margins and reproducibility checks rather than presenting it as an exact disproof.

II. MODEL AND LATE-TIME CRITERION

Consider four strictly ordered levels $E_1 < E_2 < E_3 < E_4$ and a population column vector $\boldsymbol{p}(t)$. Its autonomous dynamics is

$$\dot{\boldsymbol{p}} = M\boldsymbol{p}, \qquad M_{ij} = a_{ij} > 0 \quad (i \neq j), \\ M_{jj} = -\sum_{i \neq j} a_{ij}. \tag{1}$$

where a_{ij} is the rate for $j \to i$. A fixed bath at inverse temperature β_b imposes detailed balance,

$$a_{ij}\pi_j = a_{ji}\pi_i, \qquad \pi_i = \frac{e^{-\beta_b E_i}}{Z(\beta_b)}. \tag{2}$$

For $i < j$, we take the six downward rates $d_{ij} = a_{ij}$ as independent and set $a_{ji} = d_{ij}e^{-\beta_b(E_j - E_i)}$. Strict positivity makes the finite chain irreducible. A Gibbs preparation at inverse temperature β is denoted by $\boldsymbol{g}(\beta)$, with $g_i(\beta) = e^{-\beta E_i}/Z(\beta)$; changing β changes only the initial state, not the bath or the generator.

Let $D = \mathrm{diag}(\boldsymbol{\pi})$. Detailed balance makes $S = D^{-1/2}MD^{1/2}$ real symmetric. Order the eigenvalues as $0 = \lambda_1 > \lambda_2 > \lambda_3 \geq \lambda_4$ and assume that λ_2 is simple. If $\boldsymbol{u}_2$ is a real unit eigenvector of S, define

$$\boldsymbol{v}_2 = D^{1/2}\boldsymbol{u}_2, \qquad \boldsymbol{\ell}_2 = D^{-1/2}\boldsymbol{u}_2, \\ b_2(\beta) = \boldsymbol{\ell}_2^{\mathsf{T}}[\boldsymbol{g}(\beta) - \boldsymbol{\pi}]. \tag{3}$$

and the normalization-independent preparation amplitude $\mathcal{A}(\beta) = |b_2(\beta)|$. With this fixed normalization,

$$D_{\mathrm{KL}}[\boldsymbol{p}(t;\beta)\|\boldsymbol{\pi}] = \frac{b_2(\beta)^2}{2}e^{2\lambda_2 t} + o(e^{2\lambda_2 t}). \tag{4}$$

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TABLE I. Triplet tests for the witness. $F_{\rm app}$ is $F_>$ or $F_<$ according to the listed branch. Positive m_{ijk} means that the applicable strict necessary condition is excluded.

Triplet	κ	branch	$F_{\rm app}/\gamma$	m_{ijk}
123	70.1750	$>$	-16.25159	1.152595
124	9.97487	$>$	-0.0104988	0.00502333
134	1.02071	$>$	-0.0553119	0.0201868
234	0.0788045	$<$	-0.0417796	0.00283252

Consequently, for two cooling preparations $0 < \beta_h < \beta_c < \beta_b$, the strict reversal $\mathcal{A}(\beta_h) < \mathcal{A}(\beta_c)$ guarantees that the hotter state is closer in relative entropy at sufficiently late times, even if it starts farther away.

III. TRIPLET SCREENING PREDICATE

For an ordered triplet $i < j < k$, retain the off-diagonal rates of the full four-level generator and define

$$q = e^{-\beta_b(E_j-E_i)}, \qquad r = e^{-\beta_b(E_k-E_j)},$$
$$\kappa = \frac{a_{ik}+a_{jk}}{a_{ij}(1+q)}, \tag{5}$$
$$F_> = a_{ik}\left(1+\frac{qr}{2}\right) - a_{jk}\left(1+\frac{r}{2}\right) - a_{ij}(1-q), \tag{6}$$
$$F_< = a_{ik}q - a_{jk}. \tag{7}$$

The published three-level necessary condition is $F_> > 0$ when $\kappa \geq 1$ and $F_< > 0$ when $\kappa \leq 1$ [4]. At $\kappa = 1$ the two signs agree; none of the triplets below lies on that boundary. The proposed global screen excludes a larger system when every triplet fails its applicable strict condition. Importantly, this test uses the original internal edge rates; it is not a principal-submatrix generator with external escape terms, an effective-rate model, or an adiabatic elimination.

IV. FOUR-LEVEL WITNESS

Set $\beta_b = 1$ by measuring energies in units of β_b^{-1} and measure rates in a common inverse-time unit γ. The witness is

$$(E_1, E_2, E_3, E_4) = \left(0, \frac{1}{200}, \frac{7}{20}, 14\right), \tag{8}$$
$$(d_{12}, d_{13}, d_{23}, d_{14}, d_{24}, d_{34}) = \gamma\left(\frac{1}{10}, 1, 13, \frac{99}{100}, 1, \frac{3}{4}\right). \tag{9}$$

Table I evaluates Eqs. (5)–(7). For compact comparison we also give $m_{ijk} = -F_{\rm app}/(a_{ik}+a_{jk}+a_{ij})$. Every applicable value is negative, and the smallest normalized exclusion margin is 2.8325184×10^{-3}.

The relaxation spectrum, ordered from stationary to fastest, is

$$\frac{\boldsymbol{\lambda}}{\gamma} = (0, -1.25273315932984, -2.74000084438500, -22.8583224729113). \tag{10}$$

Thus the slow mode is simple. Its separation from the next mode, normalized by the full spectral width, is $(\lambda_2-\lambda_3)/(-\lambda_4) = 0.0650646033548$.

Now take two Gibbs preparations on the cooling branch, $\beta_h/\beta_b = 1/50$ and $\beta_c/\beta_b = 1/20$. Equation (3) gives

$$\begin{aligned}\mathcal{A}(\beta_h) &= 0.0628307488471692912,\\ \mathcal{A}(\beta_c) &= 0.0630538586374105295,\\ \mathcal{A}(\beta_c) - \mathcal{A}(\beta_h) &= 2.2310979024123833 \times 10^{-4} > 0.\end{aligned} \tag{11}$$

This is the ordering forbidden by the proposed cooling screen. The hotter state does start farther from equilibrium: $D_{\rm KL}[\boldsymbol{g}(\beta_h)\|\boldsymbol{\pi}] = 2.53001045904$, whereas $D_{\rm KL}[\boldsymbol{g}(\beta_c)\|\boldsymbol{\pi}] = 1.74010206475$. Direct propagation with the same fixed generator gives $D_h - D_c = 1.6527113 \times 10^{-10}$ at $\gamma t = 6$ and $-2.4701607 \times 10^{-14}$ at $\gamma t = 8$, so the checked distance ordering changes between those times. The small late-time difference is not the sole discriminator: the resolved amplitude inequality in Eq. (11) independently fixes the eventual ordering through Eq. (4).

V. DISCOVERY AND NUMERICAL CONTROL

The mechanism was found by a deterministic finite scan followed by constrained local optimization. That finite search is discovery provenance rather than a global certificate. After identifying a boundary mechanism—a nearly degenerate lowest pair together with a widely separated top level—we rationalized the parameters and froze Eqs. (8) and (9) before the high-precision calculation.

For verification, the generator was assembled in two separately coded conventions: an ascending-energy column generator following Eq. (1), and a descending-energy row generator mapped back by reversal and transposition. The mapped matrices agree. The detailed-balance similarity was diagonalized at 100 and 180 decimal digits, and the amplitudes, spectral gap, triplet predicates, and matrix-exponential dynamics were recalculated at both precisions. The decisive quantities agree beyond 90 decimal places. At 180 digits the largest reported matrix or modal residual is below 2×10^{-180}, compared with the 2.23×10^{-4} amplitude difference and the 2.83×10^{-3} smallest normalized predicate margin. Probability conservation, stationarity, detailed balance, positivity along the checked trajectories, left/right eigenpair equations, and biorthogonal normalization were tested explicitly.

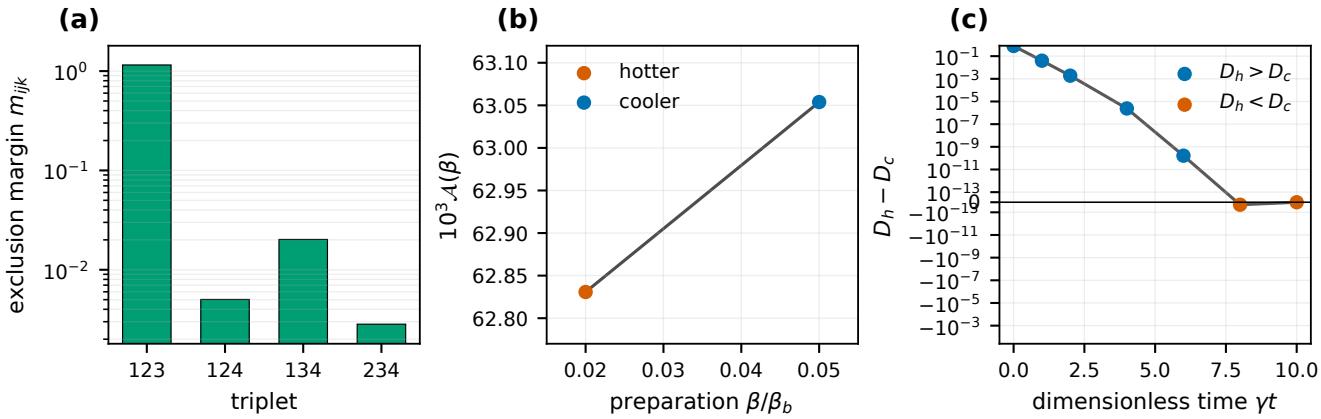


FIG. 1. Three views of the same frozen numerical witness. (a) All four normalized exclusion margins are positive. (b) The normalized slow-mode amplitude increases from the hotter preparation ($\beta/\beta_b = 0.02$) to the cooler one (0.05), reversing the proposed cooling monotonicity. (c) Relative entropy initially favors the cooler state, while the directly calculated difference $D_h - D_c$ becomes negative at the checked late times. Markers are archived high-precision outputs and connecting lines are guides to the eye. The plotting code was generated with substantive assistance from OpenAI Codex; it deterministically reads the archived machine-readable verifier output without manual data entry.

These checks control ordinary floating-point and representation errors very strongly, but arbitrary precision is not directed rounding. We have not enclosed the Gibbs factors or the isolated eigenpair with intervals and do not attach a theorem-level sign certificate to the result. The exact-rational input and large stable margins make the witness reproducible as a computational result; stronger exact or computer-assisted language would require a new enclosure argument.

A. AI-use disclosure for the computational workflow

The user designated the source conversation and required a staged verification workflow. The topic originated in an AI-generated batch on that designated page, which the user also supplied as an attachment; a local AI supervisor then selected this candidate and formalized its scope before research. OpenAI Codex using the GPT-6 Astra and GPT-5.6 Sol model families (accessed September 2026) provided substantive assistance with literature synthesis, numerical-search design, code generation and debugging, verification planning, figure-code generation, and scientific interpretation. The AI-assisted output was constrained by a frozen model statement and checked through the two generator representations, precision scaling, residual tests, and direct dynamics described above. No AI system is an author. Before submission, the named author must independently inspect the cited sources, code, outputs, figures, and claims, revise them as needed, and accept full responsibility; that human review is pending for this draft.

VI. PHYSICAL INTERPRETATION AND LIMITATIONS

The witness is not tied to an unphysical sign choice. In the chosen rate unit, take a common Arrhenius attempt rate $\Gamma = 14\gamma$. For every edge $i < j$, choose a symmetric barrier satisfying

$$\beta_b B_{ij} = \beta_b E_j + \ln(14\gamma/d_{ij}). \tag{12}$$

Because every $d_{ij} < 14\gamma$, the barrier lies above the upper level. The downward rate is $\Gamma e^{-\beta_b(B_{ij}-E_j)} = d_{ij}$ and the reverse rate is $d_{ij}e^{-\beta_b(E_j-E_i)}$, exactly as in Eq. (2). This supplies one standard ideal-bath realization of the population generator; it is not a claim of a unique or universal microscopic spectral density.

The strict predicate margins, nonzero spectral separation, and strict amplitude difference also imply continuity under sufficiently small parameter perturbations, so the effect is not confined to a tuned equality. We do not quantify that neighborhood here. The large scaled top energy, $\beta_b E_4 = 14$, lies outside the compact energy boxes used in the reported finite-temperature random tests [4, 5]; this helps explain why a finite no-hit scan does not establish the proposed implication.

The scope is deliberately narrow. We show one cooling pair for one four-state classical population generator with a simple slow mode. We do not classify inverse heating, degenerate slow subspaces, systems with $N > 4$, coherence, memory, a changing bath, transient crossings of unrelated observables, or microscopic universality. Nor does a bounded literature search establish absolute priority. Within these limits, the calculation demonstrates the physical point: local three-level necessary-condition screening can miss collective four-level anomalous cooling.

VII. CONCLUSION

A finite-temperature four-level detailed-balance generator can fail every applicable three-level necessary-condition predicate and still display a Markovian Mpemba cooling reversal. The explicit rational fixture, resolved slow mode, stable amplitude margin, and direct entropy overtaking make this conclusion reproducible at controlled numerical scope. The example therefore invalidates the proposed universal cooling screen at that scope, while leaving the heating branch, general N, and exact computer-assisted certification open.

DATA AVAILABILITY

The machine-readable high-precision output, the independent-convention verifier, the plotting script, and environment requirements accompany this manuscript. The verifier reconstructs the generator from the exact parameters in Eqs. (8) and (9); no downloaded paper or private runtime log is required to reproduce the numerical tables and figure.

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OpenAI Codex also assisted substantively with organization and drafting of this manuscript. The scope and pending author-review requirement are stated in the AI-use disclosure above.

[1] G. Teza, J. Bechhoefer, A. Lasanta, O. Raz, and M. Vucelja, Speedups in nonequilibrium thermal relaxation: Mpemba and related effects, Physics Reports **1164**, 1 (2026).

[2] Z. Lu and O. Raz, Nonequilibrium thermodynamics of the Markovian Mpemba effect and its inverse, Proceedings of the National Academy of Sciences **114**, 5083 (2017).

[3] I. Klich, O. Raz, O. Hirschberg, and M. Vucelja, Mpemba index and anomalous relaxation, Physical Review X **9**, 021060 (2019).

[4] I. Avitan, R. Factor, and D. Gelbwaser-Klimovsky, Necessary conditions for the Markovian Mpemba effect (2026), v1, March 4, 2026, arXiv:2603.04567 [cond-mat.stat-mech].

[5] D. Gelbwaser-Klimovsky, Efficient protocol for the Markovian Mpemba effect in N-level systems (2026), v1, September 9, 2026, arXiv:2609.09988 [cond-mat.stat-mech].

[6] F. Amorim, J. Wisely, N. Buckley, C. DiNardo, and D. Sadasivan, Predicting the Mpemba effect using machine learning, Physical Review E **108**, 024137 (2023).